RECOVERING AN OBJECT OF THE DERIVED CATEGORY OF A SUITABLE SUBCATEGORY AS THE DERIVED LIMIT OF ITS TRUNCATIONS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; it reduces the assertion to the companion criterion of the same section, whose hypotheses are supplied by the situation.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The situation of the Olsson-Laszlo comparison: site, sheaf of rings and weak Serre subcategory with a covering family on which the members of the subcategory are acyclic above uniform bounds. Same chapter, the situation delivered verbatim as a context span in this card (source lines 5686-5704). Supplies the hypotheses on which the statement depends.
- The companion criterion recovering an object from the derived limit of its truncations. Same chapter, the companion lemma of this section applied with the bounds supplied by the situation. Supplies the result applied in the proof.

Source reference. Original remark 0H72: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5717–5717. Navigation pointer invoked at proof line 5717; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6N: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5392–5392. Navigation pointer invoked at proof

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line 5722; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Situation 0.1. Let $\mathcal{C}$ be a site. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}$. Let $\mathcal{A} \subset \mathit{Mod}(\mathcal{O})$ be a weak Serre subcategory. We assume the following is true: there exists a subset $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ such that

(1) every object of $\mathcal{C}$ has a covering whose members are in $\mathcal{B}$, and
(2) for every $V \in \mathcal{B}$ there exists an integer d_V and a cofinal system Cov_V of coverings of V such that

$$H^p(V_i, \mathcal{F}) = 0 \text{ for } \{V_i \to V\} \in \mathrm{Cov}_V,\ p > d_V,\ \text{and } \mathcal{F} \in \mathrm{Ob}(\mathcal{A})$$

Lemma 0.2. *In Situation 0.1 for any $E \in D_{\mathcal{A}}(\mathcal{O})$ the map $E \to R\lim \tau_{\geq -n} E$ of Derived Categories, Remark 0H72 is an isomorphism in $D(\mathcal{O})$.*

Proof. Follows immediately from Lemma 0D6N. □